

Implementing and proving the Efficiency of unlimited DUORAM

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ABSTRACT

Please provide an abstract of no more than 300 words. Your abstract should explain the main contributions of your article, and should not contain any material that is not included in the main text.

Keywords: Keyword1, Keyword2, Keyword3

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1 INTRODUCTION

1.1 Roadmap

1.2 Related Work

2 TECHNICAL FOUNDATIONS

2.1 Function Secret Sharing

2.2 (Distributed) Point Functions

2.3 Distributed Comparison Functions

Grotto Insights

2.4 Beaver Triples

Multiplication / Addition Triples

Du-Attalah Protocol

3 OBLIVIOUS RANDOM ACCESS MEMORY (ORAM)

3.1 Oblivious Access and Oblivious Transfer

3.2 Oblivious Datatypes

3.3 Oblivious RAM

3.4 DUORAM

4 CONTRIBUTIONS

4.1 Target

4.2 Implementation

4.3 Grotto Framework

4.4 Testing and Benchmarking

5 IMPLEMENTATION

5.1 Target

5.2 Possibilities

Increasing the Size of the existing DPFs

CDPF Intervals

Improving Cyclic Shifting

5.3 Implementing the Possibilities

5.4 Test Results

5.5 Performance Benchmarks

6 EVALUATION

6.1 Benchmarking and Comparison to existing Implementation

6.2 Benchmarking and Comparison between the implemented Solutions

7 CONCLUSIONS

7.1 Acknowledgments

REFERENCES